Prime Multiples

Time limit: 1.00 s
Memory limit: 512 MB

CSES 2185 | Mathematics | https://cses.fi/problemset/task/2185

You are given k distinct prime numbers $a_1,a_2,\ldots,a_k$ and an integer n.

Your task is to calculate how many of the first n positive integers are divisible by at least one of the given prime numbers.

Input

The first input line has two integers n and k.

The second line has k prime numbers $a_1,a_2,\ldots,a_k$.

Output

Print one integer: the number integers within the interval $1,2,\ldots,n$ that are divisible by at least one of the prime numbers.

Constraints

- $1 \le n \le 10^{18}$
- $1 \le k \le 20$
- $2 \le a_i \le n$

Example

Input
20 2 2 5

Output
12

Explanation: the 12 numbers are 2,4,5,6,8,10,12,14,15,16,18,20.